

MongoDB Tutorial: Deleting Documents from a Collection

This tutorial explains how to delete documents from a MongoDB collection using `deleteOne()` and `deleteMany()`. It also demonstrates how to delete all documents from a collection through practical examples.

1. Overview of Document Deletion in MongoDB

MongoDB provides two primary methods for deleting documents within a collection:

1. `deleteOne(filter)` Deletes the first matching document based on the specified filter.
 2. `deleteMany(filter)` Deletes all documents that match the filter condition. Passing an empty filter `{}` removes all documents in the collection.
-

2. Syntax and Usage

2.1 deleteOne()

Purpose

Deletes the first document matching the filter.

Syntax

```
db.collectionName.deleteOne({ field: value })
```

Behavior

- If multiple documents match the filter, only the first match is deleted.
 - Typically, `_id` is preferred for deletion because it is unique.
-

2.2 deleteMany()

Purpose

Deletes all documents matching the filter.

Syntax

```
db.collectionName.deleteMany({ field: value })
```

Behavior

- Every document that satisfies the condition is removed.

2.3 Delete All Documents in a Collection

Purpose

Removes all records from the collection without dropping the collection itself.

Syntax

```
db.collectionName.deleteMany({})
```

Important

- Curly braces `{}` **must be present** but empty.
- This removes all documents but keeps the collection structure intact.

3. Practical Demonstration

The transcript uses a database named **school** and a collection named **students**.

3.1 Starting MongoDB Server

```
mongo
```

3.2 View All Databases

```
show dbs
```

3.3 Select the Database

```
use school
```

3.4 View Collections

```
show collections
```

Output included collections:

- `personal`
- `students`

3.5 View All Documents in `students`

```
db.students.find()
```

4. Deleting Documents

4.1 Delete a Document Using `_id` (Using `deleteOne`)

Example

Suppose the document for *Salman Khan* has the following `_id`:

```
{ "_id": ObjectId("65a123abc...") }
```

Command

```
db.students.deleteOne({ _id: ObjectId("65a123abc...") })
```

Result

```
{ acknowledged: true, deletedCount: 1 }
```

The document is successfully removed.

4.2 Delete a Single Document by Matching a Field (`deleteOne`)

Example: Delete the first student whose class is `"BCA"`.

```
db.students.deleteOne({ class: "BCA" })
```

Behavior

- Only the first matching record is deleted.
 - MongoDB does not delete all "BCA" students.
-

4.3 Delete Multiple Documents Using deleteMany

Example: Delete all students whose class is "BCA".

```
db.students.deleteMany({ class: "BCA" })
```

Result

```
{ acknowledged: true, deletedCount: 2 }
```

All documents with "class": "BCA" are deleted.

4.4 Delete All Documents from the Collection

Command

```
db.students.deleteMany({})
```

Note

- Curly braces are required.
- This removes all documents while keeping the collection.

Verification

```
db.students.find()
```

Output:

```
[]
```

The collection is now empty.

5. Summary of Key Commands

Delete One Document

```
db.collection.deleteOne({ field: value })
```

Delete Multiple Documents

```
db.collection.deleteMany({ field: value })
```

Delete All Documents

```
db.collection.deleteMany({})
```
